

Match the following concepts in atomic structure with the correct descriptions:

1. Concepts Description

I. Photoelectric Effect A. Region in space where there is a high probability of finding an electron

II. Bohr Model B. Explains behavior of electrons using wave properties

III. Quantum Numbers C. Quantum mechanical model principle describing electron configuration

IV. Orbitals D. Quantum model that incorporates fixed orbits for electrons

Choose the correct answer from the options given below:

(1) I-B, II-D, III-A, IV-C

(2) I-D, II-A, III-C, IV-B

(3) I-A, II-B, III-D, IV-C

(4) I-A, II-C, III-B, IV-D

Answer Key: 2

Solution:

? Photoelectric Effect: This phenomenon occurs when light (or other electromagnetic radiation) is shone on a metal's surface, causing the ejection of electrons. It does not directly involve the wave nature of electrons or quantum numbers.

? Bohr Model: Developed by Niels Bohr, this model describes electrons moving in fixed orbits around the nucleus and was a precursor to quantum mechanics.

? Quantum Numbers: These are numbers that describe values of conserved quantities in the dynamics of a quantum system. They do not directly describe electron ejection or orbital shapes.

? Orbitals: These are functions that describe the regions in space where there is a high probability of finding an electron, crucial in the quantum mechanical model of atoms.

Hence, Option (2) is the right answer.

Match the following aspects of redox reactions with their corresponding descriptions:

1. Aspects Description

I. Oxidation Number A. Technique for balancing equations where electrons gained equal electrons lost

II. Oxidation B. Increase in oxidation number

III. Reduction C. Decrease in oxidation number

IV. Balancing Redox Reactions D. Assignment and calculation rules for electrons in atoms

Choose the correct answer from the options given below:

(1) I-D, II-B, III-C, IV-A

(2) I-C, II-D, III-A, IV-B

(3) I-B, II-C, III-A, IV-D

(4) I-A, II-C, III-B, IV-D

Answer Key: 2

Solution:

? Oxidation Number: This is a conceptual charge assigned to an atom in a molecule based on electron distribution, crucial for identifying the states of oxidation and reduction.

? Oxidation: This involves the loss of electrons, which typically results in an increase in oxidation number, indicating a higher positive charge or less negative charge on the atom.

? Reduction: Conversely, this involves the gain of electrons, leading to a decrease in oxidation number, indicating a lower positive charge or more negative charge.

? Balancing Redox Reactions: This is a method used to ensure that the number of electrons lost in oxidation equals the number of electrons gained in reduction.

Hence, Option (2) is the right answer.

Match the following elements related to conic sections with their correct mathematical expressions:

1. Elements Expression

I. Parabola A. $y^2 = 4ax$

II. Ellipse B. $(x-h)^2 + (y-k)^2 = r^2$

III. Hyperbola C. $(x-h)^2/a^2 - (y-k)^2/b^2 = 1$

IV. Circle D. $(x-h)^2 = 4p(y-k)$

Choose the correct answer from the options given below:

(1) I-D, II-B, III-C, IV-A

(2) I-A, II-C, III-D, IV-B

(3) I-B, II-D, III-A, IV-C

(4) I-C, II-A, III-B, IV-D

Answer Key: 2

Solution:

? Parabola: The standard equation of a parabola oriented along the x-axis is $y^2 = 4ax$, where a is the distance from the vertex to the focus.

? Ellipse: An ellipse's standard form does not match any of the given options directly, but its general form involves the ratio of distances from any point on the ellipse to two focus points being constant.

? Hyperbola: The standard equation of a hyperbola centered at point (h, k) with transverse axis along the x-axis is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

? Circle: A circle's standard equation is $(x-h)^2 + (y-k)^2 = r^2$ where (h, k) is the center and r is the radius.

Hence, Option (2) is the right answer.